Asmi Umesh Pulgam

Tempe, AZ | asmipulgam@gmail.com | +1 (480) 457-9055 | linkedin.com/in/asmipulgam/ | asmipulgam.vercel.app/

Professional Summary

Software Engineer with programming experience building cloud-native microservices, CI/CD pipelines, and AI-powered applications, and delivering projects in teams. Skilled in LLM integration, RAG pipelines and agentic AI systems. Currently pursuing a Master of Science in Computer Science at Arizona State University (GPA 4.00/4.00) with coursework and projects in machine learning and artificial intelligence. Available for Fall 2026 and Spring 2027 co-op/internship opportunities, and full-time roles starting May 2027.

Education

Arizona State University – Tempe, AZ, US **Aug 2025 – May 2027**

Master of Science in Computer Science; GPA: 4.00 / 4.00

Savitribai Phule Pune University – Pune, MH, India **Aug 2021 – May 2025**

Bachelor of Engineering in Computer Engineering, Honors in Data Science; GPA: 3.70 / 4.00

Work Experience

Software Engineer, Infosys Finacle **Feb 2024 – Aug 2025**
Pune, India

- Transformed a monolithic banking platform into cloud-native REST microservices by migrating 15+ SOAP services with Java, Spring Boot, MongoDB, and AWS, improving API response times by 35% while reducing maintenance effort by 50%.
- Automated CI/CD and containerized deployment pipelines using Jenkins, GitLab, Docker, and Kubernetes, achieving 85% deployment automation, 60% faster releases, and 30% lower AWS infrastructure costs.

Projects

Agentic AI Financial Advisor Platform with RAG | https://sparkyfi.vercel.app/
Next.js | TypeScript | Gemini 2.5 Flash | Pinecone | Plaid API | CSS | Vercel | RAG | Agentic AI

- Developed a full-stack AI financial advisor leveraging Gemini 2.5 Flash, Plaid API, and ElevenLabs, enabling users to analyze bank statements, insurance policies, leases, and loan documents through conversational AI and voice interaction.
- Architected a production-grade RAG and agentic AI pipeline with Pinecone vector search and semantic retrieval across 2,800+ State Farm and CFPB records, delivering personalized, context-aware financial recommendations.

Natural Language-to-SQL Query Engine | https://fieldcode-e1wi-9uwl0epwk-asmipulgam-6000s-projects.vercel.app/
Python | PostgreSQL | Anthropic Claude API | SQL | LangChain | Git | MCP | Redis | Snowflake

- Engineered an AI platform powered by Anthropic Claude API, empowering 100+ non-technical users to query live PostgreSQL, MySQL, Redis, and Snowflake databases through natural language.
- Implemented secure MCP-based connectors and an interactive frontend with LangChain query caching, decreasing average query response times by 40% and expanding self-service data access.

Distributed Database Performance Optimization System | https://github.com/asmipulgam/Geo-Distributed-HealthCare-Database-System
CockroachDB | MongoDB | Docker | SQL | Python | Distributed Systems

- Designed a cross-region database platform using CockroachDB, MongoDB, Docker, and SQL, providing fault-tolerant, highly available data storage across multi-region clusters.
- Optimized query performance by refining replication, data placement, and consistency models, reducing cross-region latency by 30% and improving throughput by 25%.

Privacy-Preserving Language Model Fine-Tuning and Deployment | https://github.com/asmipulgam/Hate-Speech-Detection-Using-Small-Language-Model
Python | PyTorch | Hugging Face Transformers | TinyLlama | Qwen2.5 | LoRA | QLoRA

- Built an on-device hate speech detection system using TinyLlama, Qwen2.5, PyTorch, and Hugging Face Transformers, ensuring 100% on-device inference without transmitting sensitive user data.
- Enhanced Small Language Model training efficiency using LoRA and QLoRA, achieving 2-hour fine-tuning on a 4GB GPU and increasing offensive-content recall from 39% to 55%.

Skills

Programming Languages: Python, Java

AI/ML: TensorFlow, PyTorch, LLM, LangChain, RAG, NumPy, Pandas, Scikit-Learn, Matplotlib, Agentic AI, Fine-Tuning

Databases: MySQL, PostgreSQL, MongoDB, Redis, PySpark, Kafka, ChromaDB, Pinecone, Snowflake

Cloud Platforms: AWS (EC2, S3, Lambda, DynamoDB, SQS, ELB, CloudWatch), Docker, Kubernetes, CI/CD

Tools: Git, GitHub, Claude API, Gemini API, Vercel, VS Code, REST API, Prompt Engineering

Leadership Experience

Technical Lead, Google Developer Student Club (GDSC) **Aug 2023 – May 2024**

- Led technical workshops and coordinated hackathons and speaker sessions, guiding students through web development, cloud services, application of data structures and algorithms and AI/ML fundamentals.

Club Advisor, Association of Computer Engineering Students (ACES) **Aug 2022 – May 2025**

- Advised student leadership on technical event planning and mentorship initiatives, and guided onboarding for incoming club officers each academic year.